Uniform Positive Interpolation and Quantified Sewing for Marked Honeycomb Polygons

A local Ram analysis of sharp finite bridge moments

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Abstract

We study the finite strip functional and the marked-pair sewing interfaces of the September 26 manuscript *Uniform marked-polygon estimates and sharp finite bridge moments*. For its specific complete jet spaces, we give explicit positive translated-minor bounds valid at every width, with a uniform Taylor remainder. For the reciprocal hyperbolic denominator we isolate the slowest exponential in its Laplace representation, obtaining an explicit leading-product interpolation residual and a smaller product error. The oblique projection and the physical bridge normalization remain distinct from a Hermitian Gram solve. We also quantify the endpoint corner loss in the source's exterior sewing and retain the singular marked-separation power in a finite cutoff budget. These are local analytic certificates conditional, where stated, on the source's physical current and geometric identifications. Contribution assessment: C3/5, explicit all-width positivity and residual-product certificates together with quantified geometric cutoff losses refine specific finite interfaces, without independently establishing the global marked-polygon or bridge-exponent claims.

1 Physical law and proof boundary

Honeycomb vertices are triangle centers in the unit equilateral tiling; paths start and end at boundary-edge midpoints. Their length ℓ counts visited centers, and their critical weight is

$$w(\gamma) = \kappa^{\ell(\gamma)}, \qquad \kappa = (2\cos(\pi/8))^{-1}.$$

There are $\ell - 1$ full internal links and two terminal half-links. The source's strict bridge measure B_h has one fixed initial midpoint, any terminal midpoint on the opposite pure cut, and all centers strictly between the cuts. The Euclidean width is hd_0, $d_0 = \sqrt{3}/2$. These are unnormalized positive path weights, not a length-conditioned or infinite-path law.

The source [1] claims, uniformly at sufficiently large integral heights,

$$B_h \asymp h^{-1/4}, \qquad \sum_{\gamma \in \mathcal{B}_h} \ell(\gamma) w(\gamma) \le C h^{13/12}, \qquad B_h\{\ell \ge c h^{4/3}\} \ge c h^{-1/4}.$$

Its marked estimate has additional restrictions. For $P = P_1 e_1 + P_2 e_2$, $P_1, P_2 > 0$ integral, let $d_{\mathrm{mark}} = P_1 + P_2$ be even and $N_{\mathrm{per}} = k d_{\mathrm{mark}}$ with integer $k \ge 2$. On the cylinder of period kP, two

identical axial marks translated by P have polygon mass

$$\mathcal{K}(N_{\mathrm{per}}, d_{\mathrm{mark}}) \le C N_{\mathrm{per}}^{-4/3} (N_{\mathrm{per}}/d_{\mathrm{mark}})^{15/8}. \tag{1}$$

Marks mean actual crossed tiling edges, not independent sampling of two points. Simple unoriented polygons through the two marks are counted once. We keep the period, displacement, parity, and axis restrictions when using this inequality.

The notation d in the source's positive interpolation section instead means the number $2N+1$ of nodes for a strip of width N. We call that number m. It is not d_{mark}, and strip width is not automatically the period N_{per}.

Our all-width minor and Laplace-residual results are proved below for explicit analytic functions. Transferring them to the bridge weight uses the source's finite double-row physical identification. The uniform current expansion, nonzero denominator normalization, and actual geometric sewing required for (1) are separate imported inputs. Finite tests do not establish them.

2 The actual strip functional

Set

$$L = \frac{\pi}{8}, \quad C_m(x) = \cosh(m\sqrt{x}), \quad D(x) = 2C_L(x) - 1, \quad a_* = D(-4) = \sqrt{2} - 1.$$

The power series makes C_m entire in x, including negative x without a square-root branch choice. On $x = z^2 > 0$ put

$$\mathrm{d}\mu(x) = \frac{zD(z^2)}{\sinh(4Lz)}\,\mathrm{d}z, \qquad g(x) = \frac{\sinh(L\sqrt{x})}{\sqrt{x}},$$

and

$$M_N = \mathrm{span}\{1, x, \dots, x^{N-1}, \partial_m^j C_m|_{m=L} : 0 \le j < N\}.$$

The source's finite transfer theorem produces a scalar ρ and $f \in M_N + \mathbb{C}g$ such that

$$\mathcal{L}(H) = 4a_* H(-4) + \rho H(0) + \int H f \,\mathrm{d}\mu, \qquad 2B_N = \mathcal{L}(1/D). \tag{2}$$

Here brackets are bilinear integration, not a Hermitian inner product; $H(-4), H(0)$ must be specified and the integral absolutely convergent. The exact annihilation conditions are

$$\mathcal{L}(M_N) = 0, \quad \mathcal{L}(C_{L/2}) = 0, \quad \mathcal{L}(C_{3L/2} - C_{L/2}/D) = 0.$$

In the double row the two boundary variables commute but satisfy $e_\ell^2 = e_r^2 = 0$ separately. Loop cancellation selects a simple common eigenline at fixed width. Its eigenvalue has mixed coefficient q, and the physical translation average gives $B_N = q/2$: one transfer step contains two rows. This fixed-width rank-one limit alone supplies no width-uniform spectral gap. The bulk fusion and boundary reflection formulas preserve ordered slots and their gauge phases. A generic exact interpolation of matrix entries would not identify the physical path sum.

For real spaces define

$$c_1 = C_{L/2}, \quad c_3 = C_{3L/2}, \quad h_0 = D^{-1}, \quad h_1 = c_1/D,$$
$$F_0 = M_N + \mathbb{R}c_1, \quad F = F_0 + \mathbb{R}c_3, \quad G = M_N + \mathbb{R}g.$$

For ordered positive nodes $X = (x_1 < \cdots < x_m)$, $m = 2N+1$, the source uses the probability density proportional to $\det F_0(X) \det G(X) \prod_i \mathrm{d}\mu(x_i)$. The two determinants are generally different. The identity $g = \int_0^L C_t \,\mathrm{d}t$ and complete-jet positivity give a positive mixture, not equality of the two spaces.

3 An all-width translated-minor margin

The parameter form

$$C_{\sqrt{\beta}}(x) = \sum_{j\geq 0} \frac{\beta^j x^j}{(2j)!}$$

allows complete jets in β at repeated nonnegative labels. The change from m-jets at $L > 0$ to β-jets at L^2 is triangular and invertible. At label zero the complete jets are multiples of $1, x, \ldots, x^{N-1}$.

After those N polynomial rows are placed first, let $D_J(x)$ be the determinant of the Taylor coefficients at x of the remaining rows, at increasing derivative orders $J = (j_1 < \cdots < j_s)$ with $j_i \geq N$. Each coefficient means the derivative divided by $j_i!$. The remaining labels are $L^2/4$ once and L^2 repeated N times for F_0; for F add $9L^2/4$ once. Repeated labels mean complete confluent jets, with the same positive ordering convention throughout.

Lemma 3.1 (Derivative majorant)**.** *For either label list and every such J, $D_J(0) > 0$ and*

$$0 \leq D_J^{(r)}(0) \leq \theta_N^r D_J(0), \qquad \theta_N = \frac{(N+5/2)L^2}{2(2N+1)}, \quad r \geq 0. \tag{3}$$

Using the shorter label list only reduces the admissible constant.

Proof. For distinct positive labels the coefficient determinant at zero is the generalized power determinant divided by $\prod_i (2j_i)!$. It is positive: a combination of s ordered powers has at most $s-1$ positive zeros by division by the first power and successive Rolle arguments, and coalescing the nodes near one determines the positive sign. Complete confluent limits at the present positive labels preserve strict positivity.

Differentiating a Taylor coefficient of order j multiplies the coefficient of order $j+1$ by $j+1$. Thus $D_J' = \sum_i (j_i+1) D_{J+\mathbf{1}_i}$, with collisions zero. At zero the factorial ratio makes the multiplier on the incremented pure-power determinant $1/[2(2j_i+1)]$, at most $1/[2(2N+1)]$. The sum of those incremented power determinants is $(\sum \beta_i)$ times the original determinant. To check this, multiply its alternant expansion by $\sum \beta_i$ and regroup which exponent was increased; terms with coincident exponents cancel. Passing to confluent labels gives the same identity with multiplicities. Positivity therefore bounds $D_J'(0)/D_J(0)$ by $\sum \beta_i/[2(2N+1)] \leq \theta_N$. Every repeated derivative is a nonnegative combination of these same increasing-index determinants. Applying this bound successively proves (3). □

Theorem 3.2 (Uniform positivity and Taylor error)**.** *For every integer $N \geq 1$, every relevant J, and $0 \leq a \leq 4$,*

$$\frac{26}{61} \leq \frac{D_J(-a)}{D_J(0)} \leq \frac{96}{61}. \tag{4}$$

If $T_m(-a) = \sum_{r=0}^m (-a)^r D_J^{(r)}(0)/r!$, then

$$\frac{|D_J(-a) - T_m(-a)|}{D_J(-a)} \leq \frac{48}{13} \frac{(35/96)^{m+1}}{(m+1)!}. \tag{5}$$

These bounds are uniform in width, order set and the two specified label lists, not assertions about arbitrary jet configurations.

Proof. Entire analyticity and Lemma 3.1 give $|D_J(-a) - D_J(0)| \le (e^{a\theta_N} - 1)D_J(0)$. The fraction $(N + 5/2)/(2N + 1)$ decreases for $N \ge 1$. Using $\pi < 22/7$, hence $\pi^2 < 10$, gives

$$a\theta_N \le \frac{7\pi^2}{192} < \frac{35}{96} < 1, \qquad e^{a\theta_N} \le \frac{1}{1 - 35/96} = \frac{96}{61}.$$

Subtracting the resulting difference bound proves the lower margin $2 - 96/61 = 26/61$, and the upper follows directly. The exponential-series tail is at most $e^{a\theta_N}(a\theta_N)^{m+1}/(m + 1)!$ times $D_J(0)$. Divide by its proved lower margin to obtain (5). □

This strengthens the source's large-N translated-minor positivity to an explicit all-N statement for the actual labels and evaluation range. It does not give a lower singular value for an arbitrary evaluation matrix. Small node separation can still make such a matrix badly conditioned.

4 A leading-pole interpolation remainder

Let $S = 64/9$. The negative zeros of D are $-s_j$, where the positive numbers $(16n + 8/3)^2$, $n \in \mathbb{Z}$, are arranged in increasing order. Thus $s_1 = S$, $s_2 = 1600/9$, and $\sum_j s_j^{-1} < \infty$. The sine product, or the genus-zero product of this entire function with $D(0) = 1$, gives

$$D(x) = \prod_{j \ge 1} (1 + x/s_j).$$

For $0 \le a \le 4$, set $r_j = s_j - a$ and take independent exponential variables E_j of rates r_j. Their sum $Z = \sum_j E_j$ is finite almost surely since its mean is finite. Then, for $w \ge 0$,

$$h_0(w - a) = h_0(-a)\mathbb{E}e^{-wZ}, \qquad \mathbb{E}e^{-wZ} = \frac{D(-a)}{D(w - a)}. \tag{6}$$

Lemma 4.1 (An explicit survival bracket)**.** *Put $r = S - a$ and*

$$C_a = \frac{1}{(S - a)D'(-S)}, \qquad E_a = -\frac{1}{4D(a - 2S)}.$$

Both constants are positive and $D'(-S) = L\sin(L\sqrt{S})/\sqrt{S} = L\sqrt{3}/(2\sqrt{S})$. For every $t \ge 0$,

$$C_a e^{-rt} - E_a e^{-2rt} \le h_0(-a)\mathbb{P}(Z \ge t) \le C_a e^{-rt}. \tag{7}$$

The simpler lower bound $h_0(-a)e^{-rt}$ is also valid.

Proof. Write $Z = E_1 + V$. Since $r_2 > 2r$, the random variable V has a finite exponential moment at $2r$. Removing the first factor from the convergent product, before evaluating it, gives

$$\mathbb{E}e^{rV} = \frac{D(-a)}{rD'(-S)}, \qquad \mathbb{E}e^{2rV} = -\frac{D(-a)}{D(a - 2S)}.$$

The second identity is a moment of V, *not* of Z; $\mathbb{E}e^{2rZ}$ is infinite. The canceled product is finite up to its next pole and justifies the displayed evaluation.

Conditioning on V gives

$$\mathbb{P}(Z \ge t) = \mathbb{E}e^{rV}e^{-rt} - \mathbb{E}\big[(e^{r(V-t)} - 1)\mathbf{1}_{\{V>t\}}\big].$$

For $y \ge 0$, $e^{ry} - 1 \le e^{2ry}/4$, because $(u - 1)/u^2 \le 1/4$ for $u = e^{ry} \ge 1$. Hence the subtracted nonnegative term is at most $\frac{1}{4}e^{-2rt}\mathbb{E}e^{2rV}$. Multiplying by $h_0(-a) = 1/D(-a)$ proves (7). Finally $Z \ge E_1$ gives the simpler lower bound. □

Theorem 4.2 (Consecutive-node residual certificate)**.** *Let $m \geq 1$ and let $v_1, \ldots, v_m \geq 0$ be distinct nodes with $v_i + a > 0$, $0 \leq a \leq 4$. Interpolate h_0 at these nodes by a polynomial of degree less than m, and let $\mathcal{E}_a$ be its error at $-a$. For $q > a$ put $P_q = \prod_i (v_i + a)/(v_i + q)$. Then*

$$\max\{h_0(-a)P_S,\ C_aP_S - E_aP_{2S-a}\} \leq \mathcal{E}_a \leq C_aP_S. \tag{8}$$

Confluent positive nodes are allowed by limits with the prescribed Hermite derivatives. In particular the relative width of the second bracket, measured against C_aP_S, is

$$\frac{E_a}{C_a} \prod_i \frac{v_i + S}{v_i + 2S - a}. \tag{9}$$

Proof. Translate to $w = x + a$ and evaluate at zero. For the input e^{-tw} the error in interpolation at $w_i = v_i + a$ is $\mathbb{P}(Y \leq t)$, where Y is the independent sum of exponentials of rates w_i. This follows by partial fractions of $\prod_i w_i/(w_i + z)$: its inverse Laplace transform integrates to the same exponential combination as the Lagrange interpolation weights. Equivalently the difference has the same Laplace transform in t, and both continuous functions vanish at $t = 0$. Integrating this identity against the positive law in (6) gives $\mathcal{E}_a = h_0(-a)\mathbb{P}(Y \leq Z)$. The Laplace transform of Y is the product of its individual transforms. Taking expectations in (7) therefore gives (8), and division yields (9). Positive-node confluent limits are continuous on both sides. □

For $a = 0$ and all $v_i = 1$ in the confluent limit,

$$P_S = (9/73)^m, \qquad P_{2S} = (9/137)^m, \qquad C_0 = 2\sqrt{3}/\pi.$$

The relative product error shrinks as $(E_0/C_0)(73/137)^m$. This is an exact analytic example, not a sampled simulation of the determinant node law. For nodes very large compared with m, the product ratio can remain close to one. Thus no uniform small relative remainder over all configurations follows merely by increasing the number of nodes.

5 Transfer to the actual jet functional

The preceding theorem concerns consecutive polynomial powers. It is not an equality for the source's larger jet spaces. We now state precisely what survives the transfer. Write $e_yH = H(y) - I_{F_0,X}H(y)$ and $E_yH = H(y) - I_{F,(0,X)}H(y)$.

Proposition 5.1 (Two different node products)**.** *For $N \geq 1$, take e_{-a} with $a = 0, 4$, or E_{-a} with $a = 4$. Order its nodes $v_1 < \cdots < v_m$, so that all $v_i + a > 0$. Define P_q^{all} using all m nodes and $P_q^{[N]}$ using the smallest N nodes. Then*

$$\max\{h_0(-a)P_S^{\mathrm{all}}, C_aP_S^{\mathrm{all}} - E_aP_{2S-a}^{\mathrm{all}}\} \leq e_{-a}h_0 \text{ or } E_{-a}h_0 \leq C_aP_S^{[N]}.$$

The corresponding residuals of h_1 are nonnegative.

Proof. We spell out the source's positive-interpolation mechanism needed here. Translate by a. Place the powers $1, w, \ldots, w^{N-1}$ first in the jet basis. Cauchy–Binet expansion in Taylor coefficients expresses the normalized interpolation-error determinant as a convex combination of errors for ordered monomial systems containing these powers. The weights are the translated minors times positive ordered-power evaluation determinants, divided by their positive sum. Theorem 3.2 makes every such minor positive for every N, and the entire power series converge absolutely at the finitely many nodes.

For a completely monotone input, these monomial errors increase when any power is increased while retaining its position. Here is the sign argument underlying this classical comparison. For e^{-tw} multiply evaluation columns by e^{tw}. The coefficient matrix of $e^{tw}w^j$ is totally nonnegative: it is the exponential of the one-step shift, obtained as a limit of products of nonnegative bidiagonal matrices. Expanding ordered-power determinants then shows that the coefficient of position i, numbered from zero, in its interpolant has sign $(-1)^i$. Increasing that power changes the error by this coefficient times the new residual of the old power; the latter has the same sign by the ordered-power determinant formula. Integrating against a positive Laplace measure preserves the comparison. Its lower endpoint is the full consecutive system. Sending the last powers successively to infinity leaves the first N powers at the smallest N nodes, giving the upper endpoint. This limiting operation concerns a fixed finite node configuration, not a uniform approximation in N.

Theorem 4.2 applies to the two endpoints for h_0. For h_1, each numerator zero $-b_j$, with $b_j = ((2j+1)\pi/L)^2$, can be paired with a distinct smaller denominator zero $-s_j$. After translation, the normalized factor is

$$\frac{1+w/(b_j-a)}{1+w/(s_j-a)} = \frac{s_j-a}{b_j-a} + \left(1-\frac{s_j-a}{b_j-a}\right)\frac{s_j-a}{s_j-a+w}.$$

This is the Laplace transform of a mixture of zero and an exponential variable. The unpaired denominator factors are exponentials; summability of the reciprocal rates gives a convergent nonnegative sum. Thus $h_1(w-a)$ is completely monotone and its interpolation errors are nonnegative. □

Even though each full consecutive-node product has an explicit relative remainder, the upper and lower products in Proposition 5.1 differ. Comparing their averages requires the source's characteristic-product estimates and control of the event that its Nth node is small. Neither our finite-node theorem nor a deterministic matrix inversion supplies that probability estimate.

5.1 A nonvanishing elimination denominator

Let ν be the determinant-product probability above. Its mean interpolant is the oblique projection onto F_0 preserving pairings with G. Indeed, expand both determinants in the integral defining ν and use Fubini; the normalizing integral is the determinant of the cross-pairing matrix times $m!$. Replacing a row by H yields Cramer's rule for exactly those moment constraints. Absolute integrability holds here because the fixed jet growth is dominated by the exponential tail of μ. This is the classical determinant-integration identity, not orthogonal projection onto F_0.

Set

$$A = -\nu[e_0c_3], \quad x = \nu[e_0h_1], \quad y = \nu[e_{-4}h_0], \quad u = \nu[e_{-4}h_1], \quad v = \nu[e_0h_0].$$

Complete-jet positivity gives $e_0c_3 < 0$: append the larger label $9L^2/4$ to the label list and use the error determinant, whose sign at a point below $m = 2N+1$ nodes is negative. Consequently $A > 0$, and Proposition 5.1 gives $x, y, u, v \geq 0$. Define ν_3 by density $-e_0c_3/A$ relative to ν, and $R = \nu_3[E_{-4}h_0] > 0$. The source's exact elimination identity becomes

$$\frac{B_N}{2a_*} = \frac{AR + xy - uv}{A + x}. \tag{10}$$

In particular its denominator in the original sign convention is $-(A+x)$ and cannot vanish at *any* width $N \geq 1$. No lower bound uniform in N is asserted: obtaining the scale $A \gtrsim N^{-1/2}$ remains an asymptotic input.

Corollary 5.2 (Certified box evaluation)**.** *Suppose finite intervals for A, R, x, y, u, v are certified for the actual averages, with $A_- > 0$ and all other lower endpoints nonnegative. The minimum and maximum of the right side of (10) over this box occur among its 64 corners. Multiplication by $2a_*$ gives a rigorous enclosure for the physical strip weight under the source identification.*

Proof. Holding five variables fixed, the quotient is a linear fractional function of the sixth with positive denominator. Its derivative has constant sign, or is identically zero. Move that variable to an endpoint without increasing the value for the minimum, and repeat. The maximum is identical. □

This enclosure respects correlations only by discarding them; it can be wide or have a negative lower endpoint. Positivity of the true path mass does not justify silently modifying the moment intervals. Certified integration of ν and ν_3 is still required; the corollary is not such an integration algorithm and we do not report numerical values of B_N.

6 Endpoint corner losses without an exponent sacrifice

We next work conditionally on the source's actual two-sided exterior sewing. The paths, axial marks, and rounded hexagons are those of its sewing section. In particular the conditional gap measure is *unnormalized*: it is restricted to $r < s$ in a square of side comparable to the chord scale l, and is not renormalized after restriction. For fixed inner radius the gap $g = s - r$ has density at most C/l up to Cl, with rounded bounded-gap atomic mass at most C/l. These bounds are imported geometric inputs, not consequences of a moment or Gram identity.

Lemma 6.1 (Integrated corner cost)**.** *Let $1 \le t \le l$, and suppose a conditional sewing cost obeys*

$$K(g,t) \le C(1+\min(g,t))^{-1/2} \min\{1, t/(1+g)\}.$$

Under the preceding gap law, including its bounded-gap atom, its integral $\Xi(t,l)$ satisfies

$$\Xi(t,l) \le C' l^{-1/2} (t/l)^{1/2} [1 + \log(l/t)]. \tag{11}$$

Constants depend only on the geometric bounds just stated.

Proof. Integrate separately over $g \le t$ and $g > t$. Up to the same fixed geometric constants this is bounded by

$$\frac{1}{l}\left[2(\sqrt{1+t}-1) + \frac{t}{\sqrt{1+t}} \log_+ \frac{1+Cl}{1+t} + 1\right].$$

The last term pays the rounded atom. Since $t \ge 1$ and $t \le l$, this is at most $C\sqrt{t}\, l^{-1}[1 + \log(l/t)]$, proving the claim. □

The source replaces the square-root logarithm by $(t/l)^\eta$ for a fixed $\eta < 1/2$; retaining the displayed integral avoids that loss. It does not strengthen the underlying geometric connector estimate.

Proposition 6.2 (Dyadic corner budget)**.** *In the source's corner decomposition, suppose bins with $t/l \asymp 2^{-j}$ have at most $C(1+j)$ choices for the larger corner. Then restricting the smaller corner to $t \le \delta l$ costs a factor at most*

$$C\delta^{1/2}[1 + \log(1/\delta)]^2, \qquad 0 < \delta \le 1/2, \tag{12}$$

in its second-moment sewing bound. For a positive path measure of mass at most $CR^{3/4}$ and corresponding second moment at most the factor in (12) times $CR^{41/12}$, its discarded first moment is at most

$$C\delta^{1/4}[1 + \log(1/\delta)] R^{25/12}. \tag{13}$$

Proof. Lemma 6.1 supplies $C2^{-j/2}(1+j)$, and the larger-corner choices supply the second factor $1+j$. If $z = 2^{-1/2}$, then exactly

$$\sum_{j\ge J}(j+1)^2 z^j = z^J\left[\frac{(J+1)^2}{1-z} + \frac{2(J+1)z}{(1-z)^2} + \frac{z(1+z)}{(1-z)^3}\right].$$

Choose J within a fixed integer of $\log_2(1/\delta)$. This proves (12). Cauchy–Schwarz against the original positive, unnormalized measure gives $R^{(3/4+41/12)/2} = R^{25/12}$ and the square root of the cutoff factor, proving (13). □

For a corner cutoff expressed as $t \le \delta' R$, first retain chords $l \ge \delta_0 R$ with fixed $\delta_0 > 0$, then use $\delta = \delta'/\delta_0$. Constants can depend on δ_0. The source's short-chord removal remains a separate step. The estimate must not be used with an unconditioned, renormalized gap density or with an unrevealed adjacent facet treated as independent.

7 Retaining the marked-separation singularity

The following extension is conditional, not a replacement theorem for the source's magnetic current calculation. Suppose its same admissible pairs obey, for a fixed $q \ge 1$,

$$\mathcal{K}(N,d) \le C_q N^{-4/3}(N/d)^q. \tag{14}$$

At geometric scale $R \ge 2$, the source's construction has $O(R^2)$ first marks, $O(d)$ possible displacements at integer distance d, $d \le C_0 R$, and period $N \asymp R$. These counts already restrict to the actual axial translations; enlarging the sum to all positive integers only increases it.

Proposition 7.1 (Finite ratio transition)**.** *Under these hypotheses the pair sum is bounded by*

$$C_q' R^{2/3+q} H_q(M), \quad H_q(M) = \sum_{d=1}^{M} d^{1-q}, \quad M = \max(1, \lfloor C_0 R \rfloor). \tag{15}$$

For $q \ge 1$ the explicit bound

$$H_q(M) \le \begin{cases} 1 + (M^{2-q}-1)/(2-q), & q \ne 2, \\ 1 + \log M, & q = 2 \end{cases}$$

is valid. Consequently, multiplying by the source's $CR^{3/4}$ exterior sewing factor gives respectively

$$C_q' R^{41/12}\left(1 + \frac{1}{2-q}\right)\ (1 \le q < 2), \quad C_2' R^{41/12}(1+\log R), \quad C_q' R^{17/12+q}\left(1 + \frac{1}{q-2}\right)\ (q > 2). \tag{16}$$

Comparability constants are uniform only when those in (14) and the geometry are uniform; powers of their fixed scale ratios may depend on q.

Proof. For each first mark multiply (14) by the number $O(d)$ of displacements. Sum and then multiply by $O(R^2)$, obtaining $R^{2-4/3+q}H_q(M)$. Since x^{1-q} is decreasing for $q \ge 1$, $H_q(M) \le 1 + \int_1^M x^{1-q}\,\mathrm{d}x$. Evaluate the integral, and use $M \le CR$ in the three cases. The additional $3/4$ in the sewing power gives $8/3 + 3/4 = 41/12$ and $2/3 + q + 3/4 = 17/12 + q$. □

At the source value $q = 15/8$, limiting the pair sum to $d \le \delta R$ gives a factor at most $C\delta^{1/8}$ relative to $R^{8/3}$, provided $\delta R \ge 1$; if it is smaller than one there are no such integer distances. This is a cutoff of a pair observable, not an assertion that whole paths containing any close pair have small mass. The source's coincident and axial exceptional terms, bounded separately in its sewing, must still be added, including its $O(R^{37/12})$ term.

For fixed $q < 2$, the small-distance contribution tends to zero like δ^{2-q} with a constant deteriorating as $q \uparrow 2$. At $q = 2$ the logarithm prevents this uniform small-fraction conclusion. If, independently, a positive measure has first moment at least $cR^{25/12}$ and the relevant second moment obeys (16), Cauchy–Schwarz gives ordinary mass lower bounds of orders $R^{3/4}$, $R^{3/4}/\log R$, and $R^{3/4}R^{2-q}$, respectively. These are only conditional moment implications. The source's retention of long paths requires additional removal of short chords, small corners and exceptional terms, then exact-height cap sewing. We do not extend that geometric theorem to $q \ge 2$.

7.1 Concrete cutoff accounting

For macroscopic chords $l \ge \delta_0 R$ with fixed δ_0, the source's second-moment scale is $R^{41/12}$. Selecting a corner cutoff $\delta = 2^{-2J}$ yields the explicit factor $C\,2^{-J}(1 + 2J\log 2)^2$ and first-moment factor $C\,2^{-J/2}(1 + 2J\log 2)$. A near-mark cutoff $\zeta = 2^{-8J}$ at $q = 15/8$ yields $C\,2^{-J}$ for the corresponding pair sum. The unknown geometric constants have not been assigned numerical values, so these formulas describe the cost of certified constants rather than a numerical physical threshold. They also show why identifying the two cutoffs would be inefficient: their scale dependence is quite different.

8 Original Ram proofs and the physical metric

Three original private sources were read afresh [5, 6, 7]. Their use here is methodological and algebraic; they do not prove the honeycomb current identification or the marked estimate.

TR250 specifies domains of the transform and distinguishes moment coordinates $m = Ec$ from function-space norms. Its quadratic Galerkin and variational-crime passage derives a finite quadratic form by inserting an actual function expansion into the original integral. This is relevant to the exact cross-pairing matrix

$$M_{ij} = \int f_i g_j \, \mathrm{d}\mu$$

in Section 5. It does not license replacing M by $\int \overline{f_i} f_j \, \mathrm{d}\mu$: the source has two different spaces and a bilinear annihilation problem. Determinants can be positive without the cross matrix being the physical Hermitian Gram matrix.

The original physical-metric chapter derives the Gram identity and its omitted-residual term. For a real finite family $\psi_1, \dots, \psi_s$ in $L^2(\mu)$ with positive Gram matrix $H_{ij} = \int \psi_i \psi_j \, \mathrm{d}\mu$, and $b_i = \int \psi_i h \, \mathrm{d}\mu$, orthogonal projection has squared norm $b^t H^{-1} b$ and

$$\|h\|^2_{L^2(\mu)} = b^t H^{-1} b + \|h - Ph\|^2_{L^2(\mu)} \,. \tag{17}$$

To verify it, solve $Hc = b$, note $h - \sum c_i \psi_i$ is orthogonal to the span, and expand the square. This classical identity is supporting discipline, not a new contribution. In particular evaluating finitely many moments of the node law does not bound the omitted probability of small nodes used by the source.

TR117 proves the transform change of variables under specified integrability assumptions. In the source's actual measure, if $\tau = z^2$, then the identity used here is

$$\int H(x) \, \mathrm{d}\mu(x) = \frac{1}{2} \int_0^\infty H(\tau) \frac{D(\tau)}{\sinh(4L\sqrt{\tau})} \, \mathrm{d}\tau. \tag{18}$$

It is a substitution in a physical integral, not an inverse of the strip transfer operator. For example, near zero its density is comparable to $\tau^{-1/2}$, while at infinity it is comparable to $\exp(-3L\sqrt{\tau})$. Thus polynomial factors and the specified cross-products are integrable, but arbitrary rapidly growing basis functions need not be. This supplies a concrete endpoint check absent from a purely formal coordinate rewrite.

As a simple obstruction, taking a positive finite Gram matrix for $1, x$ says nothing about the tail of a nonnegative random variable whose first two moments merely match a proposed model. Likewise replacing the oblique projector by an orthogonal projector changes the annihilated test space. Our minor bounds certify specified determinant signs; they are neither uniform frame bounds nor an inverse theorem on an infinite function space.

9 Source architecture and imported obligations

The source has ten scientific section files, seven figure sources, a main file, references, and a README. The full architecture is as follows; a mapped dependency is not thereby independently verified.

Module	Mathematical role and boundary
Introduction	Precise marked displacement, nesting, chord, and fixed-height bridge statements.
Strip transfer	Ordered double row, nilpotent boundary variables, loop cancellation and physical factor 1/2; finite interpolation and the integral functional.
Strip asymptotics	Positive jet spaces, oblique determinant averages, characteristic products, residual brackets and the normalizing coefficient. Our explicit local changes occur here, not in its probability asymptotics.
Periodic polynomial	Twisted vacuum, Pfaffian denominator, fusion/wheel degree constraints and physical separating polygons. Left pairing is a transpose with crossed parameters, not an arbitrary complex adjoint.
Gas	Exact residue phases, winding neutrality, contact and Laurent-zero conventions; current expansion, spatial stability and analytic nome continuation.
Saturation	A positive comparison at the complex value i, then positivity/subharmonic propagation. It first gives logarithmic exponents, not bounded-factor estimates.
Geometry	Positive contour comparisons, confined crossings, cylinder barriers, local nesting and the cut-link occupancy identity.
Uniform nesting	Edge limits along finitely many residue classes, exclusion of zero limits by saturation, hexagon radius/rounding comparisons and first-length mass.
Magnetic	Unequal background lists, exact marked insertion, ordered interval operator, strict spectral margin, source localization and uniform Pfaffian ratios.
Sewing	Same-path two-cut moments, unnormalized gap law, corner bins, actual marked-pair injection, pruning and exact-height exterior caps.

The distinctions responsible for uniformity deserve emphasis. The gas estimate initially has $o(\log m)$ errors. The uniform nesting proof uses a fixed edge buffer and classwise limits with a power

error, then saturation to exclude zero constants in every residue class. Exponent arithmetic alone would not give this bounded normalization. For unequal marked intervals, the period is logarithmic in N and the smaller core logarithmic in d. An open interval product requires a strict norm margin and controlled interface factors; a periodic trace estimate alone does not supply it. The source's depth gauge telescopes at interfaces, but this is not a general bounded-inverse gauge on the entire infinite space.

On the geometric side, the two cuts in a second moment refer to the same path and the same completion. Recoverability and bounded multiplicity are needed before applying Cauchy–Schwarz. Independent samples with matching one-cut marginals would not substitute for this coupling. Finally a bridge at every sufficiently large specified height is not implied by a height-summed or selected-scale bound. The source supplies an additional cap construction for that purpose; this report preserves it as an input.

10 Literature, scope and practical costs

A primary-source refresh on October 9, 2026 checked the publisher record of Duminil-Copin–Smirnov [2], the arXiv abstract and journal metadata of Krachun–Panagiotis [3], and Borodin's original biorthogonal-ensemble abstract [4]. The first establishes the connective constant $2\cos(\pi/8)$, not the exact bridge exponent. The second gives quantitative sub-ballisticity and polynomial decay of critical bridge partition functions; its abstract does not assert the source's sharp $1/4$ power. Biorthogonal determinant ensembles and non-symmetric kernels are classical. These external proofs were not independently audited here. The September 26 source is treated as a claimed proof with explicitly imported dependencies, not as a settled theorem merely because its files are public. This refresh is not an exhaustive novelty search.

10.1 Literal reading boundary

All of the introduction, strip-transfer, strip-asymptotics, periodic-polynomial, saturation and sewing files were literally read, as were the main file, bibliography, README and all seven figure sources. Further literal ranges were: gas 1–67 and 586–737; geometry 1–137 and 340–376; uniform-nesting 181–336; magnetic 1–25, 324–385 and 636–949. The unread complements are, respectively, gas 68–585; geometry 138–339; uniform-nesting 1–180; magnetic 26–323 and 386–635. In particular the middle current derivation and the magnetic interval-operator construction were not fully read or independently certified. Later estimates that use them remain conditional. Figure code was inspected for its conventions, not used as proof of geometric realizability.

The original private passages were TR250 lines 121–180 and 425–477, TR117 lines 210–325, and the physical-metric chapter lines 1–210. These are 439 literal lines, not a whole-corpus audit. Pre-read hashes were persisted before display; the owned ledger records displayed ranges, post-read hashes and all unread complements. The batch identity record concerns equality of captured Git-tree bytes, not live branch status or proof validity.

10.2 Advantages, costs and next steps

The translated-minor margin removes a large-width exception in one precise analytic step. It also gives an explicit relative Taylor stopping rule without searching all node configurations. The leading-pole bracket retains a source-specific coefficient and a smaller product error, where the source only needs unspecified positive constants. The denominator sign prevents a spurious singular branch

in finite elimination. The corner budget exposes the cutoff loss and the marked-ratio calculation locates its summability transition.

These benefits have costs. Determinants may remain badly conditioned despite the relative translated-minor margin. The source law has unbounded node support; large nodes can erase a uniform relative leading-pole gain. Computing a certified average needs integration tails and actual cross-pairings, not just symbolic determinant algebra. The geometric constants have not been numerically computed. No runtime improvement, physical simulation, or finite verification of arbitrary width is claimed.

Useful next steps are to combine the explicit product remainder with a quantitative node-location theorem for the actual biorthogonal ensemble; to certify moments with tail bounds in the true measure; and to extract explicit connector constants without changing the unnormalized gap law. At $q = 2$, a new pruning argument would be required before any long-bridge theorem. For multiple slow poles, partial fractions alone give signed residues: preserving a positive remainder would need a new decomposition, not a formal alternating truncation.

The owned tests accompanying this report check finite coefficient identities, positive rational Laplace examples, box extrema, cutoff sums and selected high-precision diagnostics of the actual hyperbolic function. Their exact case and assertion counts are recorded separately from floating-point diagnostics. They do not run the source's proof as software, and no independent reviewer counts are included.

11 Conclusion

The substantive local increment is an all-width translated-minor certificate, an explicit leading-pole residual bracket with its proper jet-transfer boundary, and sharper corner and separation-cutoff accounting. Classical positivity, Laplace transforms, determinant integration and Cauchy–Schwarz are the tools, not claims of historical novelty. These deductions justify C3 as a notable local contribution, independently of L3 applicability; they do not re-establish the full current expansion, uniform marked estimate, or sharp fixed-height bridge law.

References

[1] OpenAI. *Uniform marked-polygon estimates and sharp finite bridge moments.* September 26, 2026. Pinned local scientific source. Public source locator. Captured-byte provenance and literal read boundaries are recorded separately; no live-version or global-validity assertion.

[2] H. Duminil-Copin and S. Smirnov. *The connective constant of the honeycomb lattice equals* $\sqrt{2+\sqrt{2}}$. Annals of Mathematics 175 (2012), 1653–1665. Publisher record, DOI 10.4007/annals.2012.175.3.14.

[3] D. Krachun and C. Panagiotis. *Quantitative sub-ballisticity of self-avoiding walk on the hexagonal lattice.* Annals of Probability 54 (2026), 1109–1125, DOI 10.1214/24-AOP1730. arXiv:2310.17299. Abstract and primary publication metadata checked, not a full proof audit.

[4] A. Borodin. *Biorthogonal ensembles* (1998). arXiv:math/9804027. Classical context; only the abstract/version record was checked here.

[5] M. Subbarao. Original private Ram manuscript `TR-rev-1-old-tr250.tex`, RamFoundationTRs. Freshly read lines 121–180 and 425–477: domains, physical moment coordinates, quadratic forms and variational error.

[6] M. Subbarao. Original private Ram manuscript `TR-rev-1-old-tr117.tex`, RamFoundationTRs. Freshly read lines 210–325: transform definitions and change-of-variable equivalence under integrability hypotheses.

[7] M. Subbarao. RamCalBooks, volume 21, `ch01d-physical-metrics.tex`. Freshly read complete text, lines 1–210: physical Gram geometry, omitted projection residual and stability boundaries. Private originals were not uploaded or used in online queries.